# PAVEL MALOLETKOV

**Sr. Software Engineer | Industrial Automation | C#, C/C++**

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## SUMMARY


Software Engineer specializing in C# and .NET Development with 7+ years of experience building high-performance applications for industrial systems, finance, and aviation. Expertise in .NET Core, WPF, and embedded integration, with a proven track record of optimizing codebases, automating workflows, and delivering scalable solutions under tight deadlines.


## EXPERIENCE

### Senior Software Engineer - Contract

**AVEVA**

07/2024 - Present | Philadelphia, PA

- Optimized the performance of various industrial data processing systems based on the OPC UA protocol and EventHubs streaming platform, reducing latency by 15% through data processing optimization.
- Made critical architectural enhancements, leading to significant improvements in code readability, maintainability, and a 15% more transparent debugging flow.
- Enhanced an in-house hardware simulator using .NET, enabling 20% faster and more flexible debugging for OPC UA protocol base processing module.
- Delivered mission-critical features for ARM/Linux environments, ensuring compliance with project approved hardware.
- Championed CI/CD processes using Azure DevOps, reducing deployment times by 20% and facilitating consistent, action triggered test runs and code analysis.

### Senior Software Engineer - Full Time

**Prosperity Home Mortgage**

06/2021 - 07/2024 | Chantilly, VA

- Developed a WPF-based mortgage management system using .NET, streamlining workflows for 500+ underwriters and reducing manual data entry by 60%.
- Designed and implemented integrations with third-party credit scoring APIs (REST/JSON), enhancing data accuracy and improving the overall user experience for loan officers and clients.
- Spearheaded the adoption of automated testing using xUnit, reducing deployment errors by 25% and accelerating release cycles by 15%.
- Collaborated with cross-functional teams (QA, Product, DevOps) to ensure seamless delivery of features, meeting critical business deadlines.

### Software Developer - Full Time

**NICOMATIC**

12/2018 - 06/2021 | Horsham, PA

- Contributed to the design and development of a C#-based analog signal analysis library (mirroring MATLAB models) for an aviation onboard diagnostics system, enabling real-time simulations and 40% faster diagnostics.
- Contributed to C-based embedded firmware development, implementing key features such as FPGA subsystem management and data exchange with external systems via USB and WiFi.
- Designed and developed a WPF/C#/U-Boot-based tool for embedded firmware updating, ensuring data loss resilience and robustness.
- Contributed to the development of a WPF/C# app for a diagnostics system with real-time sensor data visualization, simulation, and data analysis.
- Redesigned a UART-based data exchange protocol for fast and reliable data transfer between PC and embedded systems, enhancing efficiency by 35%.

## EDUCATION

### Master's in Electronics Engineering

**Kazan State Power Engineering University**

09/2016 - 07/2018 | Kazan, Russia

### Bachelor's in Electronics Engineering

**Kazan State Power Engineering University**

09/2012 - 07/2016 | Kazan, Russia

## SKILLS

C# | C/C++ | .NET | Linux | Docker | Azure | Agile | Embedded Systems | CI/CD | Git | Communication Protocols | Zephyr RTOS | Device Tree | REST | JavaScript

## KEY ACHIEVEMENTS

**Developed Signal Analysis Library**

- Engineered an analog signal analysis library, mirroring MATLAB models, enabling a sophisticated signal propagation simulation for aviation onboard diagnostics systems.

**Lead Firmware Update Tool Development**

- Designed a WPF/C#/U-Boot-based tool for embedded firmware updates, ensuring data loss resilience and robustness during updates.

**Architecture Refinement**

Redesigned core parts of the project architecture to enable easier integration of new features, as well as improved debugging and testing flexibility.

**Real Life Systems Simulation**

Designed and developed several systems simulating real-life behavior, enhancing testability and opening up new testing horizons, reducing testing time by 10x and decreasing overlooked bugs in release builds.

## EXPERIENCE

### Software Engineer - Full Time

**NPP PROMA**

03/2017 - 04/2018 Kazan, Russia

- Contributed to the development of a C++-based closed-loop PID controller for an industrial boiler temperature management system.
- Created an automated test stand for industrial cabinets, communicating over Modbus via RS485, reducing testing time by 60% through custom C scripting.

## PROJECTS

### Open Source Signal Processing System

Solo-designed and developed a platform-agnostic signal processing system for automotive use. Which implements ADC data collection, measurement, processing and configurable data post-processing based on user-sourced formulas, including calculations involving multiple sensor values.

Hardware-independent system, the REST API enables fine-grained configuration of connected sensors and processing algorithms, while the REST accessible JavaScript configuration allows sourcing custom ADC processing scripts.

GitHub link: github.com/pavllick/EerieLeap